

Optimal Transport-based Palette Matching for Color Transfer under Concentrated Reference Distributions

CS452000 Introduction to Image Processing

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I. Problem Description

A. Background

Color transfer seeks to recolor a source image to match the aesthetic of a reference while preserving its underlying structure. Early methods such as Reinhard et al.[2] addressed this through global statistical alignment in Lab color space, while later palette-based approaches operate on compact sets of dominant colors, enabling more flexible color transfer.

B. Previous Work

Lv and Zhang (2024)[1] proposed a fully automatic palette-based color transfer pipeline that eliminates manual palette specification. The framework extracts dominant color palettes via histogram peak analysis in Lab color space, separates foreground and background using DeepLabV3+ for independent color mapping, and resolves many-to-one palette correspondences through a consistency-keeping mechanism. Despite these contributions, the authors acknowledge that performance degrades when the color distributions of the two images are highly asymmetric.

C. Motivation

When the reference palette is highly concentrated, nearest-neighbor matching collapses source colors into a narrow target gamut hence degrading the transfer into a color filter effect. We address this by replacing nearest-neighbor matching with Discrete Optimal Transport (OT-based) to seek a globally optimal, mass-aware transport plan across palette entries rather than greedy per-color assignment.

II. Methods

A. Proposed Steps

Step 1: Palette Clustering; For each image, a palette is extracted via histogram peak analysis in Lab color space, with each palette entry weighted by its corresponding pixel count.

Step 2 (Main Implementation) : Discrete Optimal Transport-based Palette Matching; Source and

reference images are first segmented into foreground and background regions via DeepLabV3+, and the extracted palettes are partitioned into the corresponding regions accordingly. OT-based palette matching is then performed independently for each region. A cost matrix is constructed from pairwise Lab distances between source and reference palette colors, with pixel membership counts serving as probability mass weights on both sides. The resulting transport plan T^* globally minimizes total transport cost while respecting mass distributions, and each source color is remapped to a mass-weighted combination of reference colors: $f(b_i) = \sum_j (T^*_{ij} / w_i) \cdot b_j$.

Step 3: Lighting Optimization; The L channel is updated via weighted blending $L' = (1-\alpha)L_S + \alpha f(L_S)$ with $\alpha = 0.3$. An optional global lighting enhancement is further applied when the reference image exhibits abnormal exposure.

III. Possible Results

- Under concentrated reference color distributions, the proposed OT-based matching is expected to produce more balanced palette correspondences than nearest-neighbor, reducing the severity of the color filter effect by better distributing source color mass across the available reference gamut.
- For separable images with distinct foreground and background, independent region-wise matching is expected to prevent cross-region color bleeding. For inseparable images, the global OT plan maintains overall color distribution coherence.

References

- [1] Chenlei Lv and Dan Zhang. *Palette-based Color Transfer between Images*. 2024. arXiv: 2405.08263 [cs.CV]. URL: <https://arxiv.org/abs/2405.08263>.
- [2] E. Reinhard et al. “Color transfer between images”. In: *IEEE Computer Graphics and Applications* 21.5 (2001), pp. 34–41. DOI: 10.1109/38.946629.